

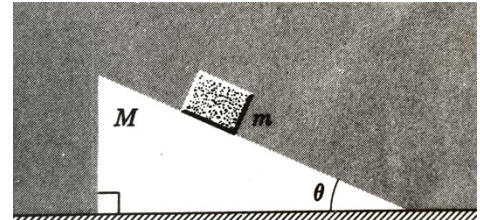
Tutorial 2: Newton's Laws of Motion (1)

Q1.

A uniform flexible chain of length l , with weight per unit length w , passes over a small frictionless, massless pulley. It is released from a rest position with a length of chain x hanging from one side and a length $l - x$ hanging from the other side. Find the acceleration a as a function of x .

Q2.

A right triangular wedge of mass M and angle θ , supporting a cubical block of mass m on its side, rests on a horizontal table, as shown. (a) What horizontal acceleration a must M have relative to the table to keep m stationary relative to the wedge, assuming frictionless contacts? (b) What horizontal force F must be applied to the system to achieve this result? (c) Suppose no force is applied to M and both surfaces are frictionless. Describe the resulting motion.



Q3.

Someone exerts a force F directly up on the axle of the pulley as shown. Consider the pulley and string to be massless and the bearing frictionless. Two bodies, m_1 of mass 1.0 kg and m_2 of mass 2.0 kg are attached, as shown, to opposite ends of the string which passes over the pulley. The body m_2 is in contact with the horizontal floor. (a) Draw a free body diagram of the pulley and for each of the masses. (b) What is the largest value the force F may have so that m_2 will remain at rest on the floor? (c) What is the tension in the string if the upward force F is 100 N? (d) With the tension determined in part (c), what is the acceleration of m_1 ?

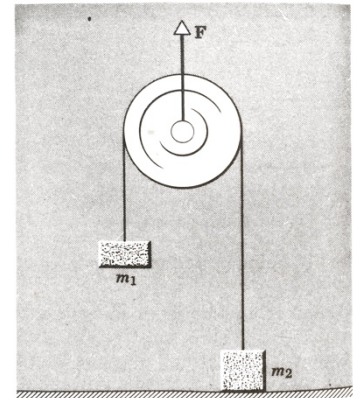


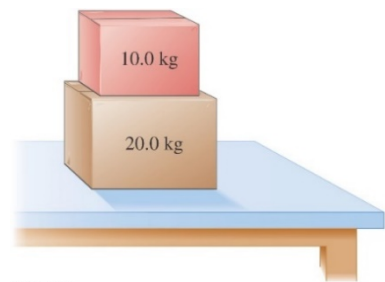
figure 5-19
Problem 27

Q4.

A block, mass m , slides down a frictionless incline making an angle θ with an elevator floor. Find its acceleration relative to the incline in the following cases. (a) Elevator descends at constant speed v . (b) Elevator ascends at constant speed v . (c) Elevator descends with acceleration a . (d) Elevator ascends with acceleration a . (e) Elevator cable breaks. (f) In part (c) above, what is the force exerted on the block by the incline?

Q5.

A 20.0-kg box rests on a table. (a) What is the weight of the box and the normal force acting on it? (b) A 10.0-kg box is placed on top of the 20.0-kg box, as shown in Fig. 4-31. Determine the normal force that the table exerts on the 20.0-kg box and the normal force that the 20.0-kg box exerts on the 10.0-kg box.



Answers:

Q1. $g(1-2x/l)$

Q2. (a) $a = g \tan \theta$ (b) $F = (M+m)g \tan \theta$

Q3. 39 N, 50 N, 40 m s⁻² upwards

Q4. (a, b) $g \sin \theta$ down the incline, (c) $(g-a) \sin \theta$ down the incline, (d) $(g+a) \sin \theta$ down the incline, (e) zero, (f) $m(g-a) \cos \theta$

Q5. (a) 196 N, 196 N, (b) 294 N, 98.0 N.